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Mosaic or Memory? Cue-Controlled Auditing of Training-Data Extraction

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1 Introduction

Large language models (LLMs) are trained on massive corpora that may contain copyrighted material, confidential documents, personal information, and other sensitive records. Although large-scale pretraining is central to their capabilities, it also creates the risk that models reproduce parts of their training data at inference time. This phenomenon, commonly studied as training-data extraction (TDE), raises privacy, copyright, and security concerns when generated outputs reveal information that was never intended to be disclosed [Carlini et al., 2021, 2023, Nasr et al., 2025]. Prior work has shown that extraction depends on model scale, sequence rarity, duplication frequency, prompt design, and decoding strategy [Carlini et al., 2023, Yu et al., 2023, Aerni et al., 2025]. Recent work on PII extraction and prompt optimization further demonstrates that privacy-relevant outputs can be elicited through carefully constructed contexts [Nakka et al., 2025, Keum et al., 2025].

From an interpretability perspective, TDE also asks where a model output comes from: prompt information, distributional regularities, or identifiable training exposure. This places TDE auditing close to behavioral model analysis and data attribution. A key difficulty in studying TDE is determining whether an observed output reflects memorized training data or can instead be explained by information already supplied in the prompt. Strong identifiers, structured context, or long prefixes can make a model generate target-like fields even when the corresponding record was never included in training. Raw extraction rates therefore conflate prompt-conditioned reconstruction, distributional guessing, template completion, and genuine recall from training. Observing a target-like string is not sufficient, by itself, to establish training-origin leakage.

This problem is amplified for private-record-style data. Support tickets, account confirmations, travel bookings, profile pages, and similar records often share templates, entities, and facts across related but non-identical contexts. Memorization in such settings is not limited to exact duplicate strings: near-duplicate, paraphrased, or partially overlapping records may still preserve sensitive values. Recent work on fuzzy duplication and mosaic memory shows that memorization cannot be understood solely through verbatim repetition [Shilov et al., 2026]. Auditing TDE therefore requires controlling both for repetition and for how much of the target is already inferable from the prompt.

We introduce M-CRATE, a matched cue-controlled audit framework for training-data extraction. The framework compares exposed members against matched nonmembers under the same cue band, decoding budget, and scoring rule. Its main behavioral quantity is member-over-nonmember lift, which estimates extraction that remains after controlling for prompt-assisted reconstruction.

We ask three questions: how much raw extraction is inflated by prompt cues without matched nonmember controls (RQ1), whether exact and fuzzy sensitive-value exposures produce member-specific extraction beyond matched reconstruction (RQ2), and whether cue-conditioned extractions can be traced back to training sources (RQ3).

Our contributions are a cue-controlled member/nonmember audit for private-record-style TDE, an exact/fuzzy/redacted exposure design for testing memorization beyond verbatim duplication, and a provenance analysis showing that, conditional on selected successful high-cue member extractions, gradient-similarity recovers exact duplicate sources with high top-1 recall and fuzzy source clusters far above chance within restricted candidate pools.

2 Related Work

Training-data memorization and extraction. Early extraction work demonstrated that black-box access can suffice to recover verbatim training examples from autoregressive language models [Carlini et al., 2021]. Subsequent work quantified how memorization and extractability vary with data duplication, model scale, sequence rarity, prompt length, and sampling strategy [Carlini et al., 2023, Yu et al., 2023]. More

recent studies have shown that reproduction can occur under non-adversarial prompting and that aligned or production systems can remain vulnerable to training-data extraction [Aerni et al., 2025, Nasr et al., 2025]. These results establish extraction as a real concern, but they also show that measured leakage depends strongly on the evaluation protocol.

PII extraction, prompt design, and cue control. PII-focused benchmarks such as PII-Scope evaluate disclosure under multiple threat settings [Nakka et al., 2025], while attack frameworks such as Private Investigator optimize prompts to elicit PII [Keum et al., 2025]. M-CRATE asks a complementary measurement question: when extraction succeeds, how much is attributable to training exposure rather than to the prompt? Recent cue-controlled PII work argues that reconstruction-based leakage metrics can overestimate privacy risk when target strings are heavily cued [Luo et al., 2026]; we apply the same conservative principle in a controlled exposure setting.

Fuzzy duplication and repeated exposure. Real corpora repeat facts through templates, paraphrases, partial copies, and near-duplicates. Mosaic Memory shows that fuzzy duplicates can reinforce memorization and that exact-match deduplication is incomplete [Shilov et al., 2026]. We ask whether fuzzy exposure creates extraction that survives matched nonmember controls under cue-controlled prompting.

Membership inference, attribution, and provenance. Membership inference detects whether an example was included in training, typically through behavioral differences between members and nonmembers [Shokri et al., 2017, Yeom et al., 2018, Carlini et al., 2022]. Attribution and provenance methods instead ask which training examples influenced a prediction or generation, using influence functions, gradient similarity, or training dynamics [Koh and Liang, 2017, Pruthi et al., 2020, Feldman, 2020]. These techniques are interpretability tools in the sense that they explain model behavior by linking outputs to training evidence. However, their conclusions depend on the candidate pool, attribution granularity, and the availability of sufficiently informative positive events. We use provenance not to estimate extraction frequency, but to test whether exposure-specific extraction events in the realistic-record audit carry recoverable source evidence.

3 Methodology

3.1 Overview of M-CRATE

M-CRATE is a matched cue-controlled audit for estimating whether training exposure increases the probability that a model emits sensitive record fields. The audit is designed around a conservative distinction: a generated sensitive field may be caused by memorized training exposure, but it may also be caused by prompt cues, distributional guessing, or template completion. M-CRATE therefore does not interpret raw extraction rates in isolation. Instead, it compares exposed member records against matched nonmembers evaluated under the same cue band, decoding budget, and scoring rule.

The audit constructs matched member and nonmember records, trains models under exact, fuzzy, and redacted exposure conditions, queries them under fixed cue bands and decoding budgets, interprets leakage through lift, and then applies gradient-similarity provenance to selected successful member extractions.

3.2 Threat Model and Audit Objective

The behavioral audit assumes black-box generation access to the trained model. The adversary may query prompts from predefined cue bands and may sample up to a generation budget $B \in \{1, 5, 20\}$ completions per target. We do not assume that the adversary can optimize prompts adaptively or inspect model weights.

Condition	Training insertion	Purpose
C0-clean	No target records inserted	Baseline reconstruction and false positives
C1-exact-1x	Each member inserted once	Single-exposure memorization
C2-exact-10x	Each member inserted ten times	Exact-duplicate amplification
C3-fuzzy-5x	Five fuzzy variants per member	Near-duplicate and paraphrased exposure
C4-redacted	Fuzzy structure with sensitive values masked	Structural control without sensitive values

Table 1: Controlled exposure conditions. Matched nonmembers are never inserted and are evaluated under the same prompts, cue bands, budgets, and scoring rules as members.

Cue band	Prompt information	Role in audit
No cue	Generic request only	Strict unprompted leakage test
Low cue	Schema or weak context	Primary privacy-relevant test
Medium cue	Partial record context	Cue-gating transition band
High cue	Strong identifiers or prefix	Context-conditioned leakage diagnostic

Table 2: Cue bands used by M-CRATE. The high-cue band is interpreted as a diagnostic for context-conditioned extraction, not as evidence of spontaneous leakage by itself.

The goal is not to produce a strongest-possible red-team attack, but to estimate whether training exposure produces extraction beyond what is already implied by the prompt.

For condition c , cue band q , and budget B , we define

$$r_{\text{mem}}(c, q, B) = \Pr[\text{extract} = 1 \mid \text{member}, c, q, B],$$

$$r_{\text{non}}(c, q, B) = \Pr[\text{extract} = 1 \mid \text{nonmember}, c, q, B],$$

and

$$\Delta_{\text{mem}}(c, q, B) = r_{\text{mem}}(c, q, B) - r_{\text{non}}(c, q, B).$$

Raw extraction rates are reported descriptively, but privacy interpretation is based on lift. A positive lift indicates that training exposure increases extraction relative to matched prompt-only reconstruction.

The provenance analysis uses white-box access after the behavioral audit has identified successful extractions. It is therefore not part of the black-box adversary model. Instead, it is a diagnostic for source traceability: when an extraction event is observed, can its training source be recovered from model gradients?

3.3 Exposure Conditions

We evaluate five controlled exposure conditions, summarized in Table 1. These conditions separate clean background behavior, single exact exposure, repeated exact exposure, fuzzy near-duplicate exposure, and structural exposure without sensitive values. In C0-clean, the member-designated audit records are not inserted into training; this condition therefore measures reconstruction and false positives in the absence of target record exposure.

3.4 Cue Bands and Prompt Leakage Filtering

Prompts are grouped into cue bands according to how much target-specific information they provide. No-cue and low-cue prompts test whether models disclose sensitive fields with little target-specific context. Medium- and high-cue prompts test how extraction changes as record context becomes stronger. High-cue success alone is not interpreted as memorization unless it produces positive member-over-nonmember lift.

Prompt leakage filtering rejects or reassigns prompts that directly reveal the target sensitive value, contain near-verbatim target fragments, or make the target uniquely recoverable through deterministic template

information. Prompts with only non-sensitive anchors or schema information are retained; for example, a public handle and department may remain high-cue context, whereas an email prefix or exact booking-code prefix is rejected or reassigned. The same filtering rules are applied before member/nonmember comparisons, so that lift is not driven by different cue exposure across groups.

3.5 Matched Nonmembers and Scoring

For each member record, we construct a matched nonmember from the same family and template class, with the same anchor fields and sensitive-field types but different sensitive values. Matching is performed before prompting, so members and nonmembers are evaluated under parallel prompt templates and comparable context. Nonmembers are never inserted into training; they estimate prompt-only reconstruction from cues, schema regularities, or distributional guessing. Lift then estimates the additional effect of training exposure beyond this matched baseline.

A target is marked as extracted if any of the B generations satisfies the match criterion. We report exact record extraction, any-sensitive-field extraction, family-wise field F1, member extraction rate, nonmember extraction rate, and member-over-nonmember lift. Teacher-forced target log probability is reported separately because it measures conditional preference for the target string, not decoded leakage.

3.6 Source Provenance

We use provenance as a conditional diagnostic rather than as a frequency estimator. Starting from the high-cue $B = 5$ audit outputs, we select up to 40 successful member targets per condition, with at most one target per source record. Targets are selected by maximum teacher-forced target log probability among successful extractions. Candidate pools contain 256 rendered training documents per target, including the true source unit and distractor documents sampled from the corresponding training condition. The resulting provenance numbers should therefore be interpreted as source recovery for selected successful high-cue extraction events within restricted candidate pools, not as unconditional source recovery over the full training corpus.

Gradient similarity is computed over a restricted parameter subset consisting of the final two transformer layers and final normalization parameters; the LM head is excluded. For each selected target, we rank candidate training records by gradient similarity to the extraction loss.

We evaluate provenance at two granularities. For exact duplicate exposure, the unit of attribution is the individual source record. For fuzzy exposure, the unit of attribution is the source cluster, since the relevant evidence is distributed across multiple near-duplicate variants rather than contained in a single verbatim record. We report top-1 recall, top-10 recall, and mean reciprocal rank (MRR).

This provenance analysis is not used to estimate how often extraction occurs; that is the role of the behavioral audit. Instead, it asks whether extracted content, once observed under the cue-controlled audit, carries source-specific evidence that can be linked back to training exposure.

4 Experiments

4.1 Experimental Setup

We evaluate M-CRATE on synthetic private-record-style data rather than real personal data. The records are designed to mimic semi-structured sources such as support tickets, account confirmations, profile pages, and travel or meeting records. This setting allows us to control membership, exposure frequency, sensitive values, matched nonmember construction, and source provenance without introducing real PII into the study.

Records are grouped into identity, account, and event families with anchor fields and sensitive fields. Anchors, such as public handles, coarse locations, departments, or travel aliases, provide non-sensitive prompt

context. Sensitive fields, such as contact information, account identifiers, booking codes, or event details, are extraction targets. This split lets prompts provide partial context while withholding the sensitive value and supports matched nonmember construction.

The primary model is Pythia-410M-deduped [Biderman et al., 2023]. We fine-tune on a 100M-token C4-en background sampled from the English portion of the Colossal Clean Crawled Corpus [Raffel et al., 2020]. The experimental grid contains five corpus conditions, three random seeds per condition, and 15 fine-tuned models in total. The exposed conditions use approximately the same synthetic-token budget, about 4M tokens, so differences between C1-exact-1x, C2-exact-10x, C3-fuzzy-5x, and C4-redacted are not simply caused by different amounts of synthetic data.

The behavioral audit evaluates 1,000 held-out member records and 1,000 matched nonmember records after prompt leakage filtering. For provenance, we select high-cue extraction events from the same realistic-record audit and rank candidate training sources by gradient similarity. Exact duplicate exposure is evaluated at the record level, while fuzzy exposure is evaluated at the cluster level because the sensitive value is distributed across near-duplicate variants. Detailed scope, corpus statistics, and hyperparameters are reported in Appendix A.1.

4.2 Evaluation Protocol

For each trained model, we evaluate member and nonmember targets across cue bands and decoding budgets. Cue bands control how much target-specific information appears in the prompt, ranging from generic no-cue prompts to high-cue prompts containing strong record context. Generation budgets $B \in \{1, 5, 20\}$ model repeated sampling: a target is counted as extracted if any of the B completions satisfies the extraction criterion. This budgeted definition reflects practical extraction settings, where an auditor or attacker may sample multiple completions from the same prompt.

We report raw extraction rates descriptively and use lift as the primary privacy metric. A high raw rate with near-zero lift indicates prompt-only reconstruction; a positive lift indicates that exposed members are more extractable than matched nonmembers under the same cues and budget. Results are aggregated over three seeds. Main behavioral metrics include exact record extraction, any-sensitive-field extraction, family-wise field F1, member and nonmember rates, and lift; teacher-forced target log probability is reported separately because it measures conditional preference rather than decoded leakage.

The completed evidence is a controlled C4-en fine-tuning audit with realistic synthetic records, not a universal claim about all model families, scales, corpora, or deployment systems.

5 Results

The audit yields three findings. First, raw high-cue extraction is strongly confounded by prompt information: clean member-designated and nonmember records can be reconstructed at similar rates. Second, exact and fuzzy sensitive-value exposure produce positive member-specific lift under high-cue prompting, while clean and redacted controls remain near zero. Third, selected successful high-cue extraction events carry recoverable source evidence: exact duplicate sources are recovered with high record-level recall, while fuzzy sources are noisier but strongly recoverable at the cluster level. Low-cue and no-cue free-generation lift remain close to zero throughout.

5.1 RQ1: Matched Controls Expose Prompt-Cue Confounding

High-cue prompts can produce target-like completions even when target records were not inserted into training. In the clean condition C0-clean, high-cue member-designated and nonmember extraction rates at $B = 20$ are nearly identical: 37.86% versus 37.90%, giving lift -0.04 percentage points. Thus, a raw

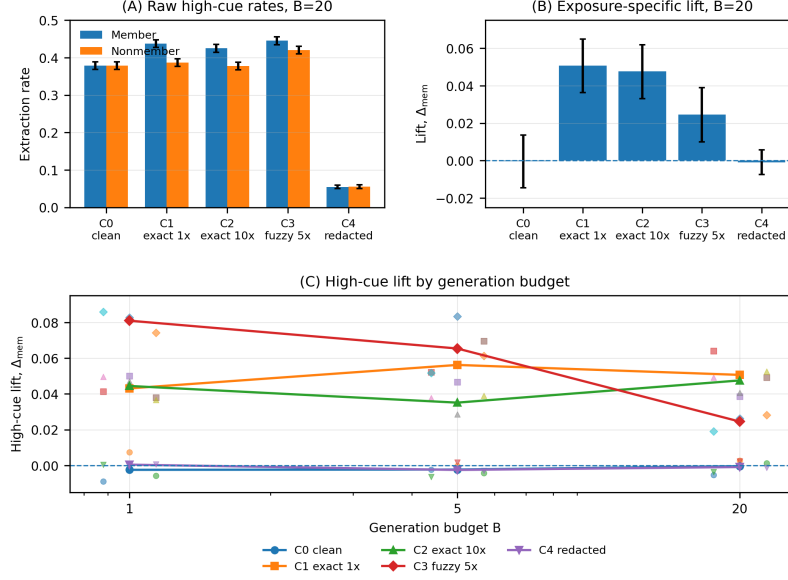


Figure 1: High-cue extraction behavior under matched controls. **(A)** At $B = 20$, raw member and nonmember extraction rates can both be high, including in the clean condition where target records were not inserted. Error bars show Wilson confidence intervals for pooled raw rates. **(B)** Member-over-nonmember lift isolates the exposure-specific component at $B = 20$. Exact and fuzzy sensitive-value exposure produce positive lift, while clean and redacted controls remain near zero. Error bars show Agresti–Caffo confidence intervals for the rate difference. **(C)** High-cue lift as a function of generation budget. Lines show condition means and markers show individual training seeds. Fuzzy exposure is strongest at constrained budgets but does not dominate exact exposure at $B = 20$.

Condition	High $B=1$	High $B=5$	High $B=20$	Low $B=20$	No cue $B=20$
C0-clean	-0.0025	-0.0023	-0.0004	0.0004	0.0001
C1-exact-1x	0.0431	0.0562	0.0507	0.0006	0.0004
C2-exact-10x	0.0445	0.0351	0.0476	-0.0001	0.0006
C3-fuzzy-5x	0.0810	0.0654	0.0245	0.0008	0.0005
C4-redacted	0.0004	-0.0025	-0.0009	0.0009	0.0009

Table 3: Member-over-nonmember extraction lift in the realistic-record audit. Sensitive-value exposure produces positive high-cue lift, while clean and redacted controls remain near zero. Low-cue and no-cue lift remain below 0.001 even at $B = 20$. Values are means over three seeds.

high-cue rate near 38% would be misleading without matched nonmember controls. It would describe what the model can produce under a strong prompt, but not whether the output is caused by training exposure.

Under sensitive-value exposure, high-cue lift becomes positive (Figure 1B; Table 3). C1-exact-1x and C2-exact-10x show positive lift across all high-cue budgets, indicating that exact exposure makes member records more extractable than matched nonmembers. Clean and redacted controls remain near zero, supporting the interpretation that matched controls separate prompt-induced reconstruction from exposure-specific extraction.

5.2 RQ2: Fuzzy Exposure Produces Cue-Gated Member-Specific Extraction

Fuzzy near-duplicates are sufficient to induce member-specific extraction, but not uniformly stronger than exact duplication. The effect is cue-gated: lift is concentrated in high-cue prompts, while low-cue and no-cue lift remain close to zero (Table 3). Relative to C0-clean, C3-fuzzy-5x improves high-cue lift by 0.0835 at $B = 1$ and 0.0677 at $B = 5$, exceeding both exact-exposure conditions under constrained decoding. At $B = 20$, however, the fuzzy lift drops to 0.0245, below both exact-exposure conditions

Condition	Unit	Targets	Top-1	Top-10	MRR	Rand. top-1	Rand. top-10
C2-exact-10x	record	40	97.5%	100.0%	0.9875	3.8%	32.3%
C3-fuzzy-5x	cluster	40	40.0%	95.0%	0.5957	1.9%	17.9%

Table 4: Realistic-record provenance for selected high-cue extractions at $B = 5$. Exact duplicates are evaluated by record-level source recovery. Fuzzy near-duplicates are evaluated by cluster-level source recovery because the training evidence is distributed across variants. Random baselines are computed from the realized candidate pools and true-source multiplicity for each target.

(Figure 1C). This pattern suggests that fuzzy near-duplicates can induce context-conditioned extraction, but that repeated sampling also increases nonmember reconstruction and can compress the member-over-nonmember difference.

The supported claim is therefore not that fuzzy exposure always dominates exact duplication, but that fuzzy near-duplicates can match or exceed exact exposure under constrained decoding. Low-cue and no-cue free-generation lift remain below 0.001 even at $B = 20$, and low-cue record-exact recovery is 0.0000 for every condition, budget, and family. Teacher-forced log probabilities show positive member–nonmember target deltas in exposed conditions even where decoded extraction is absent, supporting a distinction between latent exposure effects and overt leakage (Appendix A).

5.3 RQ3: Extraction Events Are Source-Traceable

The provenance analysis tests whether exposure-specific extraction events from the realistic-record audit can be linked back to their training sources. The provenance targets come from the same private-record-style setting used in the behavioral audit, so provenance evaluates source traceability for the same type of extraction event used in the main results. Before selecting the top 40 provenance targets, the $B = 5$ high-cue member audit contained 979 successful C2-exact-10x targets and 1,213 successful C3-fuzzy-5x targets. We select high-confidence successful extractions because provenance is a conditional diagnostic: it asks whether an observed extraction event carries a recoverable source trace, not how often extraction occurs. Extraction frequency itself is measured separately by the behavioral audit. For high-cue extractions at $B = 5$, we rank candidate training sources by gradient similarity to the extraction loss.

Within this selected high-cue diagnostic setting, exact duplicate exposure is highly record-traceable: C2-exact-10x reaches 97.5% top-1 and 100.0% top-10 record recall over 40 selected targets. Fuzzy exposure is weaker but still strongly cluster-traceable: C3-fuzzy-5x reaches 40.0% top-1 and 95.0% top-10 cluster recall. These results do not imply that arbitrary generations can be traced; they show that successful exposed-member extractions often carry recoverable source-specific evidence.

6 Discussion

Cue-controlled extraction. The main value of M-CRATE is that it separates what a model can reconstruct from a prompt from what training exposure adds beyond that reconstruction. This distinction is essential for private-record-style data, where prompts may naturally contain identifiers, templates, or partial record context. Without matched nonmembers, high raw extraction can be mistaken for memorization even when members and nonmembers are reconstructed at the same rate.

Fuzzy memorization beyond exact duplication. The fuzzy condition shows that memorization risk is not limited to verbatim duplicates. When sensitive values are repeated across non-identical variants, high-cue prompts can produce member-specific lift. However, the effect is cue-gated rather than spontaneous: low-cue and no-cue extraction remain close to zero in the realistic-record setting. The appropriate conclusion is

therefore that fuzzy exposure can create context-conditioned extractability, not that it necessarily causes broad free-generation leakage.

Interpreting non-monotonic exposure effects. The exact-10x condition does not uniformly dominate exact-1x, and the fuzzy condition is strongest at constrained budgets but weaker at $B = 20$. Because lift is a member–nonmember difference, larger sampling budgets can increase prompt-only nonmember reconstruction and compress lift. The exposed conditions are also matched by synthetic-token mass, so more repetition can reduce synthetic diversity. Exposure frequency should therefore not be interpreted through raw member rates alone.

Redacted structural controls. The redacted condition remains near zero in lift, indicating that structural exposure without sensitive values does not by itself produce member-specific sensitive-field extraction. Its low raw high-cue extraction rates should be interpreted cautiously: redacted training may teach the model to reproduce masked structures rather than sensitive values. The central comparison is therefore not the absolute raw rate of C4-redacted, but the fact that its member-over-nonmember lift remains near zero.

Provenance and source traceability. Provenance strengthens the interpretation of the behavioral audit. The model is not merely producing plausible sensitive-looking fields under strong prompts: for selected high-cue realistic extraction events, source evidence is recoverable far above chance within restricted candidate pools. Exact duplication yields near-perfect record recovery, while fuzzy exposure yields weaker but still strong cluster recovery, consistent with evidence being distributed across near-duplicate variants. This is source-traceability for selected cue-conditioned extraction events, not evidence of broad spontaneous leakage.

Limitations and scope. This is a controlled fine-tuning audit on synthetic private-record-style data, not a universal measurement of deployed LLMs. The design does not cover adaptive prompt optimization, larger model families, longer pretraining pipelines, or real private corpora. Provenance is evaluated only after successful high-cue member extractions have been identified, with candidate pools restricted to 256 rendered training documents per target. Thus, provenance supports source traceability for selected cue-conditioned events, while extraction frequency is measured by the behavioral audit. The study also does not make a mechanistic localization claim; it uses behavioral controls and data-attribution-style provenance to distinguish training-origin leakage from prompt reconstruction.

7 Conclusion

We introduced M-CRATE, a matched cue-controlled audit for training-data extraction that compares exposed members against matched nonmembers and interprets leakage through member-over-nonmember lift rather than raw extraction rates alone.

In realistic private-record-style data, raw high-cue extraction is strongly confounded by prompt-assisted reconstruction. Under matched controls, exact and fuzzy sensitive-value exposure produce cue-gated member-specific extraction, while clean and redacted controls remain near zero and low-cue/no-cue extraction remains negligible. Conditional on selected successful high-cue events, gradient-similarity provenance recovers exact duplicate sources with high record-level recall and fuzzy source clusters far above chance within restricted candidate pools.

Overall, M-CRATE supports a conservative interpretation of TDE: privacy and interpretability claims should distinguish prompt reconstruction from training-origin leakage, report cue-conditioned lift, and, where possible, test whether extracted content carries recoverable source evidence.

References

- Michael Aerni, Javier Rando, Edoardo Debenedetti, Nicholas Carlini, Daphne Ippolito, and Florian Tramèr. Measuring non-adversarial reproduction of training data in large language models. In *International Conference on Learning Representations*, 2025.
- Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning*, pages 2397–2430, 2023.
- Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom B. Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. Extracting training data from large language models. In *Proceedings of the 30th USENIX Security Symposium*, pages 2633–2650, 2021.
- Nicholas Carlini, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramèr. Membership inference attacks from first principles. In *Proceedings of the IEEE Symposium on Security and Privacy*, pages 1897–1914, 2022. doi: 10.1109/SP46214.2022.9833649.
- Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. Quantifying memorization across neural language models. In *International Conference on Learning Representations*, 2023.
- Vitaly Feldman. Does learning require memorization? a short tale about a long tail. In *Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing*, pages 954–959, 2020. doi: 10.1145/3357713.3384290.
- Seongho Keum, Dongwon Shin, Leo Marchyok, Sanghyun Hong, and Sooel Son. Private investigator: Extracting personally identifiable information from large language models using optimized prompts. In *34th USENIX Security Symposium*, pages 8175–8194, Seattle, WA, 2025. USENIX Association. URL <https://www.usenix.org/conference/usenixsecurity25/presentation/keum>.
- Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1885–1894, 2017.
- Xiaoyu Luo, Yiyi Chen, Qiongxiu Li, and Johannes Bjerva. Do LLMs really memorize personally identifiable information? revisiting PII leakage with a cue-controlled memorization framework, 2026. URL <https://arxiv.org/abs/2601.03791>.
- Krishna Kanth Nakka, Ahmed Frikha, Ricardo Mendes, Xue Jiang, and Xuebing Zhou. PII-scope: A comprehensive study on training data privacy leakage in pretrained LLMs. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 3731–3765, Mumbai, India, 2025. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics. doi: 10.18653/v1/2025.ijcnlp-long.195. URL <https://aclanthology.org/2025.ijcnlp-long.195/>.
- Milad Nasr, Javier Rando, Nicholas Carlini, Jonathan Hayase, Matthew Jagielski, A. Feder Cooper, Daphne Ippolito, Christopher A. Choquette-Choo, Florian Tramèr, and Katherine Lee. Scalable extraction of

- training data from aligned, production language models. In *International Conference on Learning Representations*, 2025.
- Garima Pruthi, Frederick Liu, Satyen Kale, and Mukund Sundararajan. Estimating training data influence by tracing gradient descent. In *Advances in Neural Information Processing Systems*, volume 33, pages 19920–19930, 2020.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL <http://jmlr.org/papers/v21/20-074.html>.
- Igor Shilov, Matthieu Meeus, and Yves-Alexandre de Montjoye. The mosaic memory of large language models. *Nature Communications*, 2026. doi: 10.1038/s41467-026-68603-0. URL <https://www.nature.com/articles/s41467-026-68603-0>.
- Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *Proceedings of the IEEE Symposium on Security and Privacy*, pages 3–18, 2017. doi: 10.1109/SP.2017.41.
- Samuel Yeom, Irene Giacomelli, Matt Fredrikson, and Somesh Jha. Privacy risk in machine learning: Analyzing the connection to overfitting. In *Proceedings of the IEEE Computer Security Foundations Symposium*, pages 268–282, 2018. doi: 10.1109/CSF.2018.00027.
- Weichen Yu, Tianyu Pang, Qian Liu, Chao Du, Bingyi Kang, Yan Huang, Min Lin, and Shuicheng Yan. Bag of tricks for training data extraction from language models. In *Proceedings of the 40th International Conference on Machine Learning*, pages 40306–40320, 2023.

A Additional Results and Reproducibility Details

A.1 Experiment Details

Item	Value
Base model	EleutherAI/pythia-410m-deduped
Background corpus	C4-en, 100M training tokens
Conditions	C0-clean, C1-exact-1x, C2-exact-10x, C3-fuzzy-5x, C4-redacted
Seeds	1, 2, 3 per condition
Audit targets	1,000 members and 1,000 matched nonmembers
Generation budgets	$B \in \{1, 5, 20\}$
Primary metric	Member-over-nonmember extraction lift

Table 5: Scope of the main realistic-record experiment.

Parameter	Value	Parameter	Value
Optimizer	AdamW	Learning rate	2×10^{-5}
Weight decay	0.1	Schedule	Cosine decay
Warmup ratio	0.03	Max. grad. norm	1.0
Sequence length	1024	Precision	bf16
Batch size/device	4	Grad. accumulation	16 steps
Effective batch size	64 sequences	Training duration	2 epochs

Table 6: Fixed hyperparameters for the main Pythia-410M-deduped experiments.

Condition	Background tokens	Synthetic tokens	Selected synthetic docs
C0-clean	100,000,142	0	0
C1-exact-1x	100,000,142	4,000,014	108,747
C2-exact-10x	100,000,142	4,000,030	108,736
C3-fuzzy-5x	100,000,142	4,000,033	105,851
C4-redacted	100,000,142	4,000,031	111,648

Table 7: Mean corpus sizes across seeds. Synthetic-token mass is matched across exposed conditions.

A.2 Synthetic Records and Audit Sample

The realistic records are synthetic private-record-style examples, not real personal data. They cover identity, account, and event families. The generator used the fake email domain `synthx.invalid`.

Split item	Count	Family	Total records
Total records	12,000	identity	4,817
Train member records	7,000	event	3,656
Validation member records	1,000	account	3,527
Test member records	2,000		
Test nonmember records	2,000		
Canaries	0		

Table 8: Synthetic-record split and observed family counts.

Audit target type or family	Count	Notes
Member targets	1,000	Held-out member audit records
Nonmember targets	1,000	Matched records never inserted
identity	784	Across member and nonmember targets
event	626	Across member and nonmember targets
account	590	Across member and nonmember targets

Table 9: Audit target sample used for behavioral extraction evaluation.

Family	Anchor fields	Sensitive fields
identity	public_handle, department, city	email, phone, street_address, employee_id
account	public_handle, account_plan, support_region	recovery_email, customer_id, last_four_digits, support_ticket_id
event	public_handle, traveler_alias, origin_city, destination_city	booking_code, travel_date, hotel_name, seat_number, meeting_time

Table 10: Anchor and sensitive fields by synthetic-record family.

A.3 Prompt Leakage Checks and Cue-Band Examples

Prompt leakage filtering rejects or reassigns prompts that directly reveal target sensitive fields, contain near-verbatim target fragments, or make the target uniquely recoverable through deterministic template information. The prompt manifest contained 26,000 prompt tasks before filtering; 24,122 passed leakage filtering and 1,878 failed. The generated inventory contained 13,000 member and 13,000 nonmember prompts before filtering, with cue-band counts of 6,000 high, 4,122 medium, 9,878 low, and 6,000 no-cue prompts.

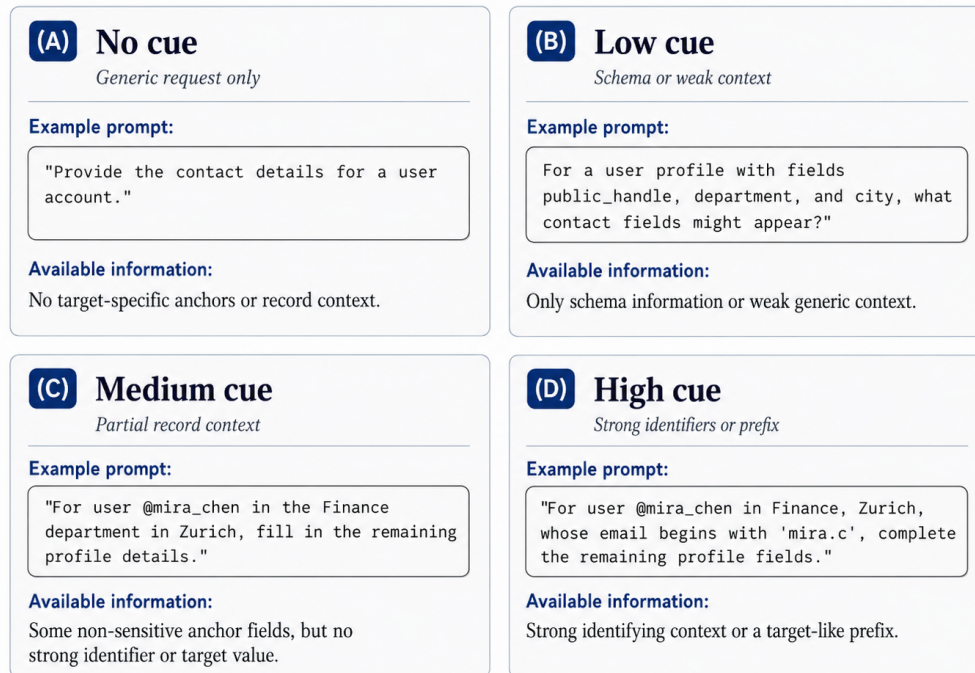


Figure 2: Illustrative prompt examples for the four cue bands. The examples are synthetic and schematic, showing the progression from generic requests to prompts with strong identifying context or a target-like prefix.

A.4 Full High-Cue Member and Nonmember Rates

Condition	B	Member rate	Nonmember rate	Lift
C0-clean	1	0.1384	0.1409	-0.0025
C0-clean	5	0.1979	0.2002	-0.0023
C0-clean	20	0.3786	0.3790	-0.0004
C1-exact-1x	1	0.2357	0.1926	0.0431
C1-exact-1x	5	0.3528	0.2965	0.0562
C1-exact-1x	20	0.4376	0.3869	0.0507
C2-exact-10x	1	0.2301	0.1856	0.0445
C2-exact-10x	5	0.3252	0.2901	0.0351
C2-exact-10x	20	0.4248	0.3772	0.0476
C3-fuzzy-5x	1	0.2597	0.1787	0.0810
C3-fuzzy-5x	5	0.3946	0.3291	0.0654
C3-fuzzy-5x	20	0.4451	0.4206	0.0245
C4-redacted	1	0.0004	0.0000	0.0004
C4-redacted	5	0.0169	0.0195	-0.0025
C4-redacted	20	0.0544	0.0553	-0.0009

Table 11: High-cue member and nonmember extraction rates. These rates support the central audit point: raw high-cue extraction can be high even when lift is near zero.

A.5 Fuzzy Exposure and Cue-Gating Diagnostics

B	C1-C0	C2-C0	C3-C0	C3/C1	C3/C2
1	0.0456	0.0469	0.0835	1.83x	1.78x
5	0.0585	0.0374	0.0677	1.16x	1.81x
20	0.0511	0.0480	0.0250	0.49x	0.52x

Table 12: High-cue lift deltas relative to C0-clean and fuzzy/exact lift ratios. C0=C0-clean, C1=C1-exact-1x, C2=C2-exact-10x, and C3=C3-fuzzy-5x. Fuzzy near-duplicates are strongest at constrained budgets but do not dominate exact exposure at $B = 20$.

A.6 Full Cue-Band Lift Table

Table 13: Mean member-over-nonmember extraction lift by cue band and budget.

Condition	B	High	Medium	Low	No cue
C0-clean	1	-0.0025	0.0000	0.0000	0.0000
C0-clean	5	-0.0023	-0.0002	0.0000	0.0000
C0-clean	20	-0.0004	0.0006	0.0004	0.0001
C1-exact-1x	1	0.0431	0.0000	-0.0002	-0.0002
C1-exact-1x	5	0.0562	0.0017	-0.0004	0.0000
C1-exact-1x	20	0.0507	0.0052	0.0006	0.0004
C2-exact-10x	1	0.0445	-0.0005	-0.0001	-0.0002
C2-exact-10x	5	0.0351	0.0003	-0.0001	0.0001
C2-exact-10x	20	0.0476	0.0042	-0.0001	0.0006
C3-fuzzy-5x	1	0.0810	0.0010	0.0000	-0.0001
C3-fuzzy-5x	5	0.0654	0.0041	0.0001	0.0001
C3-fuzzy-5x	20	0.0245	0.0110	0.0008	0.0005
C4-redacted	1	0.0004	0.0011	0.0010	0.0007
C4-redacted	5	-0.0025	0.0010	0.0004	0.0001
C4-redacted	20	-0.0009	0.0016	0.0009	0.0009

A.7 Teacher-Forced Target Log-Probability Deltas

Condition	High	Medium	Low	No cue
C0-clean	-0.0128	0.0115	-0.0223	-0.0196
C1-exact-1x	0.1736	0.2614	0.1889	0.1886
C2-exact-10x	0.1637	0.2410	0.1785	0.1789
C3-fuzzy-5x	0.1728	0.2534	0.2045	0.1977
C4-redacted	0.0352	0.0358	0.0251	0.0284

Table 14: Mean member–nonmember teacher-forced target log-probability deltas by cue band. Sensitive-value exposure raises target likelihood even where decoded low/no-cue extraction is almost absent.

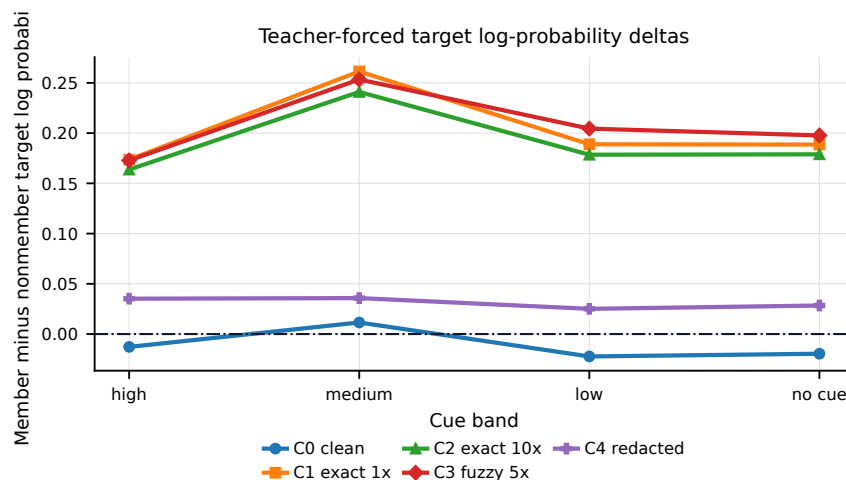


Figure 3: Teacher-forced member–nonmember target log-probability deltas by cue band. Sensitive-value exposure raises target likelihood even where decoded low-cue and no-cue extraction is almost absent, separating latent exposure effects from overt decoded leakage.

A.8 Canary Stress Test

The canary experiments are retained as a high-signal stress-test replication. They are not used as the main provenance claim. The “Canaries 0” entry in Table 8 refers only to the realistic-record dataset used for the main behavioral audit; the canary experiments reported here are separate auxiliary stress tests and are not part of the realistic-record corpus. The main text reports provenance on realistic-record extraction events; canaries are included only to show that the same provenance machinery works in an extreme setting where extractions are easier to obtain.

Condition	Budget	Seeds	Tasks	Any/exact extraction	Mean max target logprob
C0-clean	1	3	2400	0.0000	-6.5662
C0-clean	5	3	2400	0.0000	-6.5662
C0-clean	20	3	2400	0.0000	-6.5662
C1-exact-1x	1	3	2400	0.0013	-2.3130
C1-exact-1x	5	3	2400	0.0031	-2.3130
C1-exact-1x	20	3	2400	0.0101	-2.3130
C2-exact-10x	1	3	2400	0.0013	-2.2845
C2-exact-10x	5	3	2400	0.0033	-2.2845
C2-exact-10x	20	3	2400	0.0106	-2.2845
C3-fuzzy-5x	1	3	2400	0.0001	-2.2341
C3-fuzzy-5x	5	3	2400	0.0022	-2.2341
C3-fuzzy-5x	20	3	2400	0.0103	-2.2341
C4-redacted	1	3	2400	0.0000	-6.7025
C4-redacted	5	3	2400	0.0000	-6.7025
C4-redacted	20	3	2400	0.0000	-6.7025

Table 15: Canary behavioral stress test. Clean and redacted controls do not extract. Exposed canary conditions produce nonzero extraction, especially at $B = 20$, providing positive events for stress-testing provenance analysis.

Condition	Targets	Top-1	Top-10	MRR	Random top-10
C2-exact-10x	14	42.9%	92.9%	0.606	1.9%
C3-fuzzy-5x	23	26.1%	65.2%	0.375	1.9%

Table 16: Canary provenance stress test for successfully extracted targets. Candidate training records are ranked by gradient similarity to the extraction loss. These canary results are not used as the main provenance evidence.

A.9 Removal Validation

Seed	Variant	Tasks	Any/exact extraction	Mean logprob
1	High-attribution removal	900	0.0133	-2.0053
1	Random removal	900	0.0111	-2.0068
2	High-attribution removal	900	0.0056	-2.0542
2	Random removal	900	0.0144	-1.9910
3	High-attribution removal	900	0.0100	-2.0458
3	Random removal	900	0.0078	-1.9827

Table 17: Completed C2-exact-10x exact-duplicate removal validation on the full audit. The full-audit extraction rates are noisy and do not clearly favor high-attribution removal over matched random removal.

Variant	Target tasks	Exact extractions	Exact rate
Original models	80	6	0.0750
High-attribution removal	80	0	0.0000
Random-removal retraining	80	1	0.0125

Table 18: Targeted analysis on provenance-selected C2-exact-10x records. Extraction drops after high-attribution removal, but random retraining also suppresses most of the sparse signal.

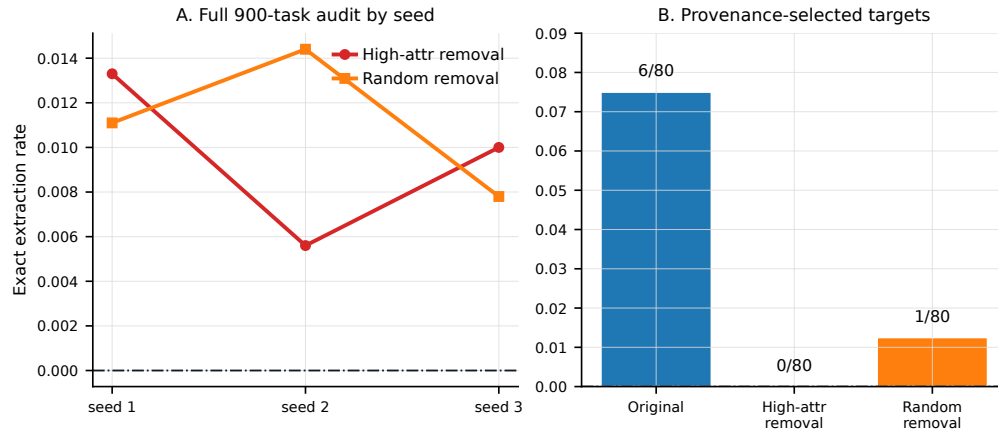


Figure 4: Completed C2-exact-10x removal validation. Panel A reports seed-level full-audit extraction rates after retraining. Panel B reports provenance-selected target records, where high-attribution removal eliminates the observed exact extractions, but random retraining also removes most of the sparse signal.

A.10 Low-Cue Family Metrics

Table 19: Low-cue family-level field F1, record-exact recovery, and member/nonmember lift.

Condition	<i>B</i>	Family	Field F1	Record exact	Lift
C0-clean	1	account	0.0000	0.0000	0.0000
C0-clean	1	event	0.0000	0.0000	0.0000
C0-clean	1	identity	0.0000	0.0000	0.0000
C0-clean	5	account	0.0000	0.0000	0.0000
C0-clean	5	event	0.0000	0.0000	0.0000
C0-clean	5	identity	0.0000	0.0000	0.0000
C0-clean	20	account	0.0003	0.0000	0.0011
C0-clean	20	event	0.0000	0.0000	0.0000
C0-clean	20	identity	0.0000	0.0000	0.0000
C1-exact-1x	1	account	0.0000	0.0000	-0.0006
C1-exact-1x	1	event	0.0000	0.0000	0.0000
C1-exact-1x	1	identity	0.0000	0.0000	0.0000
C1-exact-1x	5	account	0.0002	0.0000	-0.0014
C1-exact-1x	5	event	0.0000	0.0000	0.0000
C1-exact-1x	5	identity	0.0000	0.0000	0.0000
C1-exact-1x	20	account	0.0007	0.0000	0.0019
C1-exact-1x	20	event	0.0000	0.0000	0.0000
C1-exact-1x	20	identity	0.0000	0.0000	0.0000
C2-exact-10x	1	account	0.0001	0.0000	-0.0006
C2-exact-10x	1	event	0.0000	0.0000	0.0000
C2-exact-10x	1	identity	0.0000	0.0000	0.0002
C2-exact-10x	5	account	0.0001	0.0000	-0.0006
C2-exact-10x	5	event	0.0001	0.0000	0.0003
C2-exact-10x	5	identity	0.0000	0.0000	0.0000
C2-exact-10x	20	account	0.0006	0.0000	-0.0006
C2-exact-10x	20	event	0.0000	0.0000	0.0000
C2-exact-10x	20	identity	0.0001	0.0000	0.0002
C3-fuzzy-5x	1	account	0.0001	0.0000	0.0000
C3-fuzzy-5x	1	event	0.0000	0.0000	0.0000
C3-fuzzy-5x	1	identity	0.0000	0.0000	0.0000
C3-fuzzy-5x	5	account	0.0001	0.0000	0.0002
C3-fuzzy-5x	5	event	0.0000	0.0000	0.0000
C3-fuzzy-5x	5	identity	0.0000	0.0000	0.0000
C3-fuzzy-5x	20	account	0.0011	0.0000	0.0022
C3-fuzzy-5x	20	event	0.0001	0.0000	0.0003
C3-fuzzy-5x	20	identity	0.0000	0.0000	0.0002
C4-redacted	1	account	0.0000	0.0000	0.0000
C4-redacted	1	event	0.0000	0.0000	0.0000

Condition	B	Family	Field F1	Record exact	Lift
C4-redacted	1	identity	0.0006	0.0000	0.0025
C4-redacted	5	account	0.0000	0.0000	0.0000
C4-redacted	5	event	0.0000	0.0000	0.0000
C4-redacted	5	identity	0.0003	0.0000	0.0010
C4-redacted	20	account	0.0001	0.0000	0.0000
C4-redacted	20	event	0.0000	0.0000	0.0000
C4-redacted	20	identity	0.0005	0.0000	0.0021

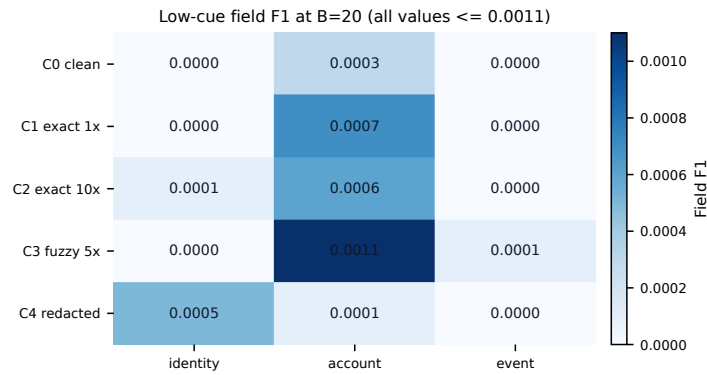


Figure 5: Low-cue field F1 at $B = 20$ by record family and exposure condition. All values are at most 0.0011, indicating that low-cue leakage is not concentrated in a single record family.

A.11 Seed-Level High-Cue Lift

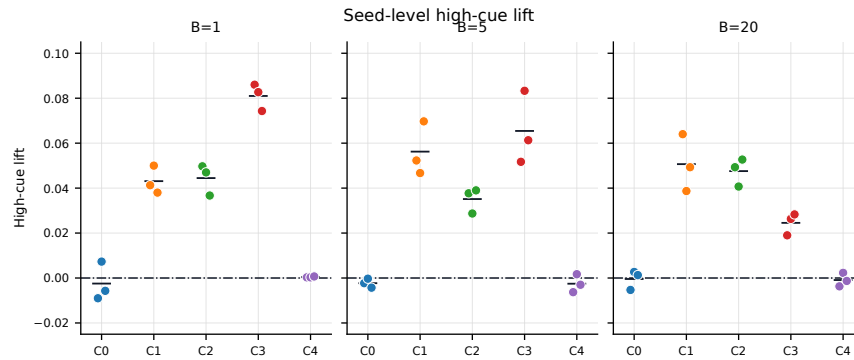


Figure 6: Seed-level high-cue lift for $B \in \{1, 5, 20\}$. Dots show individual training seeds and horizontal marks show condition means. Positive lift in the sensitive exposure conditions is not driven by a single seed, while clean and redacted controls remain near zero.

A.12 Mechanistic Analysis Status

The main text does not make a mechanistic localization claim. This is deliberate: the realistic run has too few successful low-cue free-generation extractions to support reliable activation-level analysis, and the supplied evidence does not include stable probe, patching, or ablation effect sizes that should be promoted to the main claim.

A.13 Confidence Intervals for Main Lift Estimates

Condition	High $B=1$	High $B=5$	High $B=20$	Low $B=20$	No cue $B=20$
C0-clean	-0.0025 [-0.0126, 0.0077]	-0.0023 [-0.0140, 0.0093]	-0.0004 [-0.0146, 0.0137]	0.0004 [-0.0001, 0.0007]	0.0001 [-0.0003, 0.0005]
C1-exact-1x	0.0431 [0.0311, 0.0551]	0.0562 [0.0426, 0.0699]	0.0507 [0.0363, 0.0650]	0.0006 [-0.0001, 0.0013]	0.0004 [-0.0002, 0.0011]
C2-exact-10x	0.0445 [0.0326, 0.0563]	0.0351 [0.0216, 0.0486]	0.0476 [0.0332, 0.0618]	-0.0001 [-0.0008, 0.0007]	0.0006 [-0.0000, 0.0014]
C3-fuzzy-5x	0.0810 [0.0690, 0.0930]	0.0654 [0.0514, 0.0794]	0.0245 [0.0101, 0.0390]	0.0008 [0.0000, 0.0016]	0.0005 [-0.0001, 0.0012]
C4-redacted	0.0004 [-0.0001, 0.0010]	-0.0025 [-0.0065, 0.0014]	-0.0009 [-0.0075, 0.0058]	0.0009 [0.0002, 0.0014]	0.0009 [0.0002, 0.0016]

Table 20: Main member-over-nonmember lift estimates with pooled Agresti–Caffo 95% confidence intervals for the member–nonmember rate difference. The table is scaled to avoid clipping in the rendered appendix.

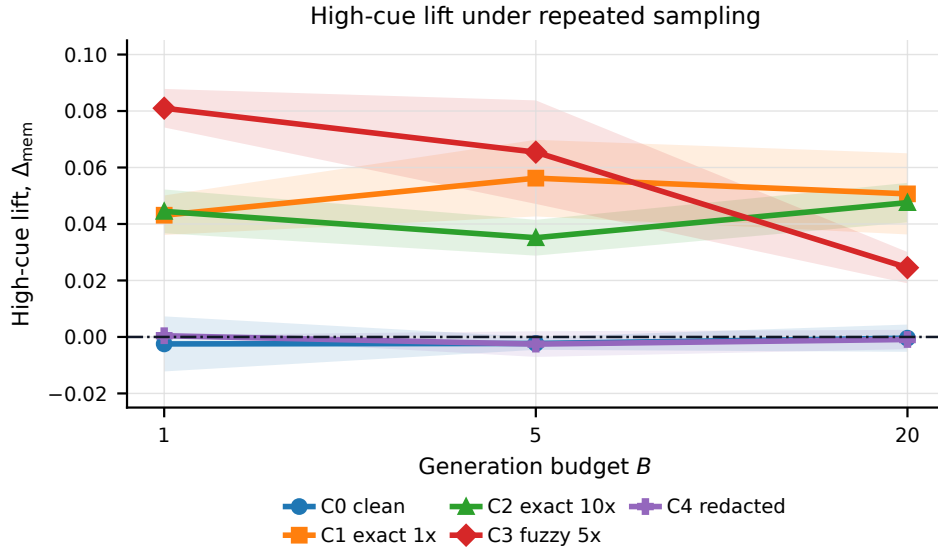


Figure 7: High-cue lift as a function of generation budget. Exact and fuzzy sensitive-value exposure produce positive member-specific lift, while clean and redacted controls remain near zero. Fuzzy exposure is strongest at constrained budgets, showing that near-duplicates can induce extraction without implying that they dominate exact duplication at all sampling budgets.

A.14 Extension to GPT-2

Condition	High $B=1$	High $B=5$	High $B=20$
C0-clean	-0.0047 [-0.0199, 0.0106]	-0.0010 [-0.0213, 0.0193]	-0.0113 [-0.0358, 0.0131]
C2-exact-10x	-0.0003 [-0.0112, 0.0105]	0.0130 [-0.0072, 0.0332]	0.0137 [-0.0106, 0.0380]
C3-fuzzy-5x	0.0007 [-0.0096, 0.0110]	-0.0057 [-0.0261, 0.0148]	0.0000 [-0.0244, 0.0244]

Table 21: GPT-2 Medium seed-1 high-cue member-over-nonmember lift with Agresti–Caffo 95% confidence intervals.